

6. There are two methods of producing ATP in the body:

- aerobic respiration
- anaerobic respiration

However, at the start of exercise, the body will use ATP that is already available in small amounts in muscle cells.

(a) Other than ATP, state the products of aerobic and anaerobic respiration. [3]

Aerobic: and

Anaerobic:

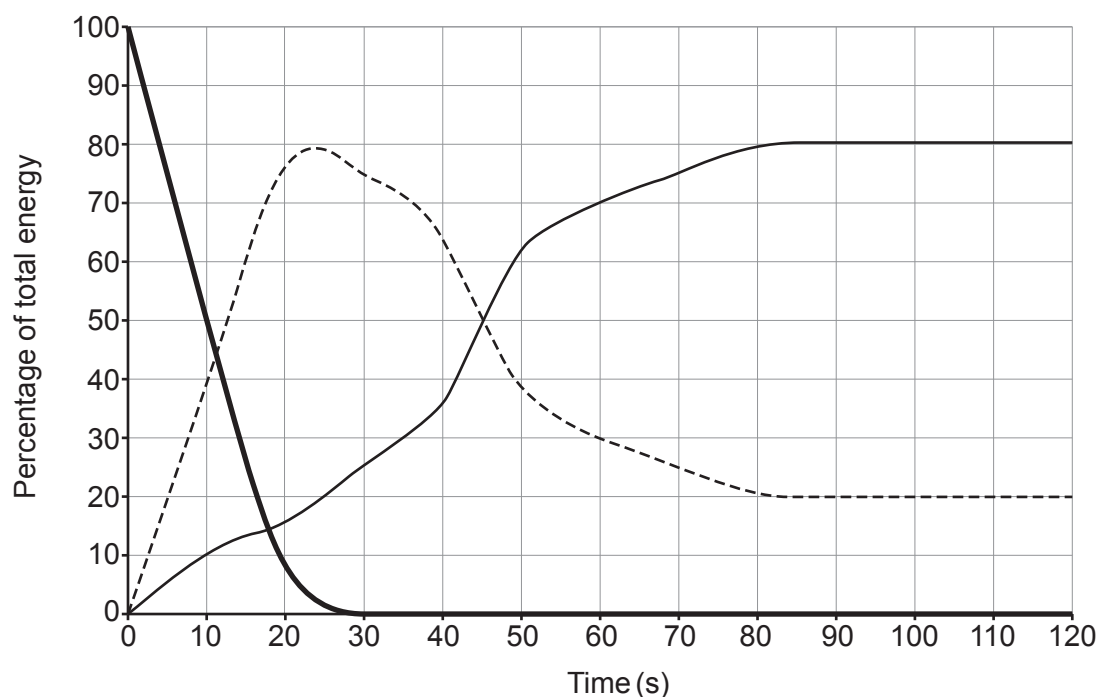
(b) State why anaerobic respiration is not suitable as a source of energy for continuous exercise. [1]

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(c) **Graph 6.1** shows how the source of ATP for muscle contraction changes over a two-minute period of exercise in an athlete.

Graph 6.1



Key: — ATP already available in small amounts in muscle cells
 - - - - - ATP from anaerobic respiration
 — ATP from aerobic respiration



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Table 6.2 shows information about ATP in muscle cells.

Table 6.2

	ATP already available in muscle cells	ATP produced from aerobic respiration	ATP produced from anaerobic respiration
Type of exercise used for	high-intensity exercise	continuous exercise	high-intensity exercise
Duration of ATP source	0–25 seconds	hours	15 seconds–2 minutes
Molecules of ATP produced per molecule of glucose		38	2

(i) Use **Graph 6.1** to suggest why it is important that muscle cells have a small amount of ATP available at the start of exercise. [1]

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(ii) Use the data from **Table 6.2** to compare the efficiency of aerobic and anaerobic respiration. Give the reason for the difference in efficiency. [2]

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